

Thymalin

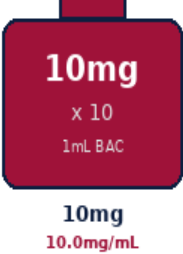
Thymic Polypeptide Extract | 10mg x 10 Vials

Immune Restoration | Anti-Aging | Thymic Regeneration | IM / SubQ

Purity: 99.8% | Endotoxin-Free | Research Use Only

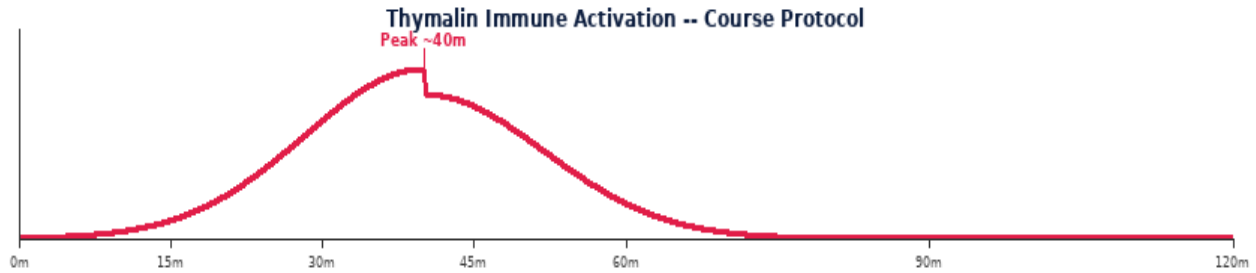
Thymalin is a polypeptide complex extracted from bovine thymus gland. Unlike synthetic Thymosin Alpha-1, Thymalin contains the full spectrum of thymic bioregulatory peptides. It stimulates T-cell differentiation, restores thymic function (which involutes with age), and normalises immune balance. Developed in Russia with clinical use since the 1970s.

Thymalin Vial Comparison



Parameter	Specification
Class	Thymic polypeptide extract -- broad thymic bioregulator
Vial Size	10mg x 10 vials Purity 99.8%
Reconstitution	1 mL BAC Water -> 10mg/mL (10,000 mcg/mL)
Standard Dose	1,000-10,000 mcg (1-10mg) IM or SubQ
Route	IM (deltoid/gluteal) preferred SubQ acceptable
Frequency	Daily x 5-10 days per course 1-2 courses per year
Cycle & Storage	5-10 day courses, 6 months between 2-8 deg C 28 days

MECHANISM OF ACTION & RESEARCH BENEFITS



Pathway	Mechanism	Benefit
T-Cell Differentiation	Promotes T-lymphocyte precursor maturation into functional T-cells in thymus; restores CD4+/CD8+ ratio; normalises T-cell receptor diversity; reverses age-related T-cell decline	Immune restoration; T-cell research; immunosenescence
Thymic Regeneration	Thymus involutes progressively after puberty; Thymalin peptides signal thymic stromal cells to regenerate; increases naive T-cell output; restores youthful immune repertoire diversity	Thymic regeneration; anti-aging immunity; longevity research
NK & Innate Immunity	Activates NK cell cytotoxicity; enhances macrophage phagocytosis; normalises innate/adaptive immune balance; antioxidant; reduces age-related immune dysfunction markers	Anti-aging; immune restoration; infection resistance

Benefit	Context
Full Thymic Complex	Thymalin contains the complete spectrum of thymic bioregulatory peptides -- broader activity than synthetic single-peptide alternatives like Thymosin Alpha-1.
Anti-Aging Immunity	As the thymus involutes with age, T-cell output declines. Thymalin course therapy stimulates thymic regeneration and T-cell renewal.
Course Protocol	Traditional clinical use: 5-10mg/day x 5-10 days, 1-2 courses per year. This simple protocol has decades of Russian clinical research.

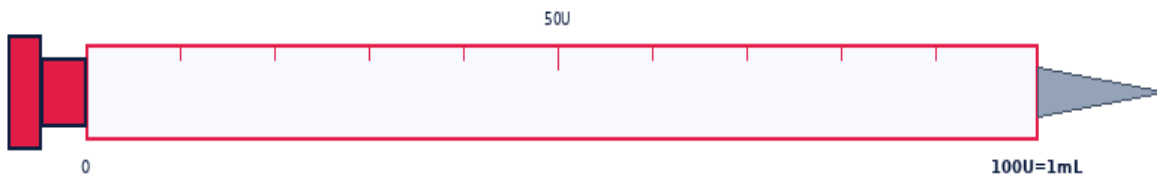
RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



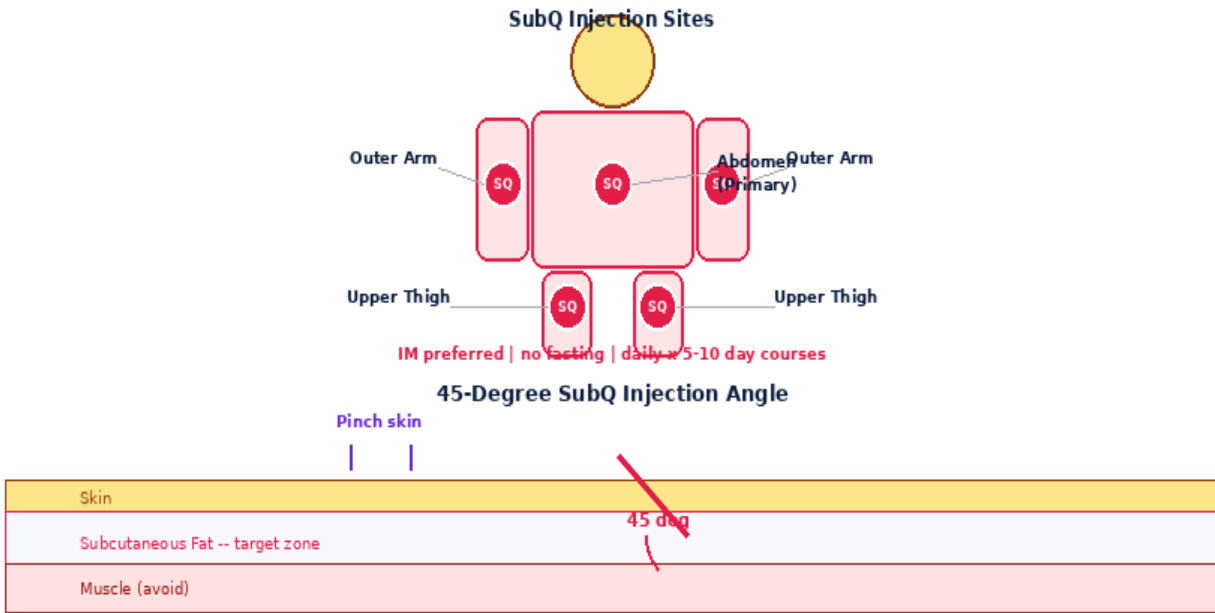
1mL Insulin Syringe | 29-31G | SubQ

Formula: $\text{Vol(mL)} = \frac{\text{Dose(mcg)}}{\text{Conc(mcg/mL)}}$ | **Units:** $\text{Vol} \times 100$

10mg x 10 -- 1mL BAC -> 10.0mg/mL (10,000mcg/mL)

Dose(mcg)	1000mcg	2000mcg	5000mcg	10000mcg
Vol(mL)	0.1000	0.2000	0.5000	1.0000
Units	10.0U	20.0U	50.0U	100.0U
Doses/Vial	10	5	2	1

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES



8-Point Rotation Map -- Rotate Every Injection



Step	Action	Detail
1	Select route	IM preferred (deltoid or gluteal). SubQ abdomen acceptable.
2	Draw dose	Standard: 1,000-5,000 mcg per injection. No fasting.
3	Inject IM/SubQ	IM: 23-25G x 1-1.5in, 90 deg, aspirate. SubQ: 29-31G, 45 deg, rotate.
4	Complete course	Run daily for 5-10 days. Record course dates.

Protocol	Dose	Frequency	Duration	Notes
Standard Course	1,000-5,000 mcg	Daily x 5-10 days	1-2 courses per year	IM preferred. Classic clinical protocol.
Anti-Aging Protocol	5,000-10,000 mcg	Daily x 10 days	1-2 courses per year	Higher research dose for longevity protocols.

Side Effect	Likelihood	Management
IM site pain	Common	Expected for IM. Rotate deltoid/gluteal.
Fatigue / immune response	Occasional	Expected during course. Rest well.

Contraindications: Cancer history, pregnancy, under 18.

1. 1 mL BAC Water -> 10mg/mL (10,000 mcg/mL).

2. No fasting required. Inject any time.
3. IM route (deltoid or gluteal) preferred for Thymalin.
4. Standard clinical protocol: 1-5mg daily x 5-10 days per course.
5. Limit to 1-2 courses per year with at least 6 months between courses.
6. Thymalin contains full-spectrum thymic peptides -- broader than synthetic Thymosin Alpha-1.
7. Store 2-8 deg C. Use within 28 days. Never freeze.
8. Monitor for immune activation signs during course (mild fever expected).
9. Not for use in autoimmune disease (immune activation risk) without physician guidance.

DISCLAIMER:

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